

Local Violence and Leader Favorability in Mali

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1 Question and approach

Is violence in a respondent's commune associated with how favorably they view the national leader? Violence is the number of ACLED events per commune and survey wave. Favorability comes from three survey waves, each with its own stratified, clustered sample and weights.

The report compares four models. Each step changes **one thing**, so any difference between neighbouring models has a single explanation:

Model	Violence	Clustering	Weights	Shows
M1	Region-wave	PSU (design)	Yes	Original specification
M2	Commune-wave	None	No	Gain from finer exposure
M3	Commune-wave	Random intercept	No	Multilevel alternative
M4	Commune-wave	Commune (design)	Yes	Main model

All four models share region and wave dummies and the same respondent covariates.

2 Data checks

The analysis file has 5,050 respondents (319 dropped for missing data) in 419 communes across 8 regions and 3 waves.

Table 1: Commune name matching. Communes are matched by name within region. A sampled commune missing from the violence file is assigned 0 events, so spelling mismatches would silently become zeros.

Check	n	Examples
Survey communes	419	
Survey communes not in official list (likely spelling)	0	
Violence communes not in official list (likely spelling)	0	
Survey communes with no violence record (set to 0 events)	18	bamako communeiii, bamako communeiv, bamako communev, bamako communevi
Survey communes whose name is ambiguous within the key	4	koulikoro diedougou, segou somo, koulikoro benkadi, sikasso kapala

Table 2: Diagnostics computed before any model is fit. K: exposure units. A: share of violence variation absorbed by region and wave dummies; whatever is absorbed cannot inform the violence estimate. P: PSUs per commune-wave. W: within-commune share of violence variation.

Di- ag- nos- tic	Meaning	Value	Status
K	Commune-wave cells in the analysis sample (share with at least one event)	687 (41%)	OK
A, re- gion	Share of region-wave violence explained by region + wave dummies (Model 1)	98%	Prob- lem
A, commune	Share of commune-wave violence explained by region + wave dummies (Models 2 to 4)	9%	OK
P	Median PSUs per commune-wave cell (share of cells with more than one PSU)	1 (10%)	Com- mune ~ PSU
W	Within-commune share of violence variance (communes surveyed 2+ waves; share of respondents)	27% (205 communes; 74%)	OK

Why Model 1 cannot work. Region and wave dummies explain 98% of the variation in region-wave violence. Model 1 must estimate the violence effect from the remaining 2%, spread over only 24 region-wave values. Measured at the commune level, the same dummies explain only 9% of the variation, leaving most of it to inform the estimate. Figure 1 shows why: communes within the same region and wave experience very different levels of violence, and Model 1 gives all of them one number.

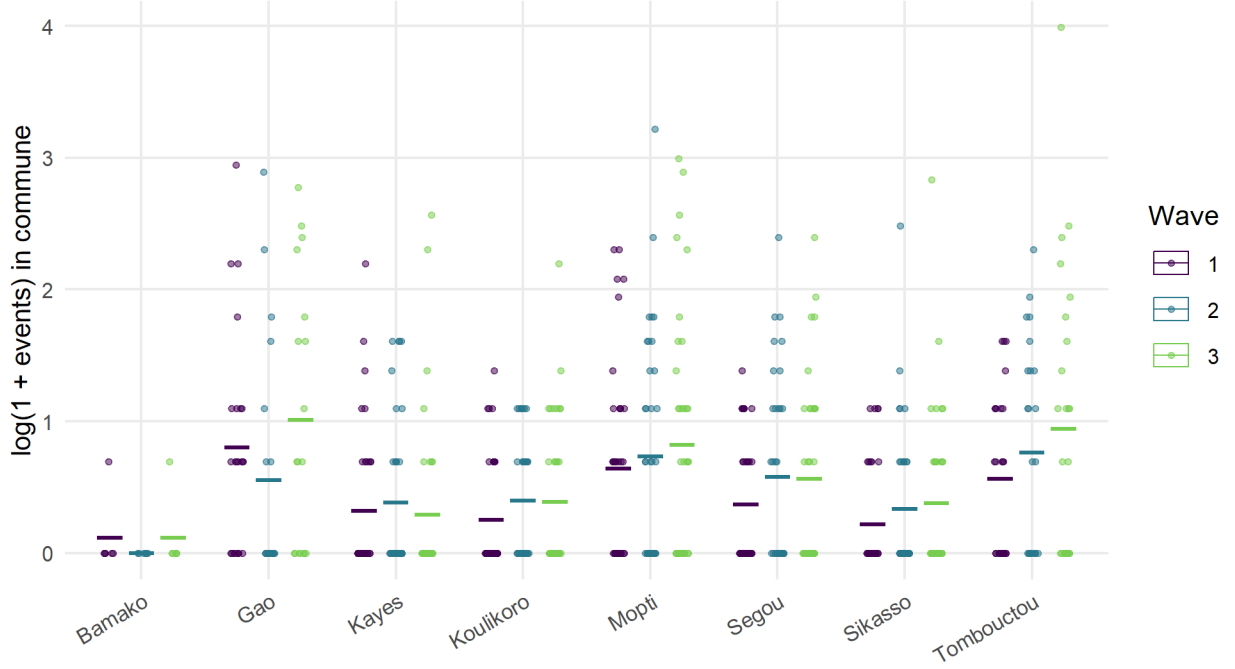


Figure 1: Commune violence within each region and wave. Each dot is a sampled commune; the bar is the region-wave average. Model 1 replaces every dot with one value per region-wave.

3 Models

3.1 Exposure

For commune c in wave w , E_{cw} is the ACLED event count and $V_{cw} = \log(1 + E_{cw})$. The log compresses a skewed count in which most commune-waves have few or no events. For Model 1, events are summed over all communes in the region: $V_{rw} = \log(1 + \sum_{c \in r} E_{cw})$.

Models 2 to 4 split commune violence into two parts (Mundlak, 1978; Bell & Jones, 2015):

$$V_c^- = \frac{1}{T_c} \sum_{w \in S_c} V_{cw}, \quad V_{cw}^{\text{within}} = V_{cw} - V_c^-$$

where S_c is the set of waves in which commune c was surveyed and T_c is how many there are. V_c^- is the commune's usual level of violence; V_{cw}^{within} is how much more or less violent that wave was than usual. A commune surveyed only once has $V_{cw}^{\text{within}} = 0$ and contributes nothing to the within estimate.

3.2 Equations

All models share the same base, with region and wave dummies and respondent covariates $\mathbf{x}_i$ (female, age per decade, ethnicity, education, urban):

$$\eta_i = \beta_0 + \sum_{r \neq \text{Bamako}} \alpha_r \text{Region}_r + \sum_{w=2}^3 \gamma_w \text{Wave}_w + \mathbf{x}_i' \boldsymbol{\beta}$$

Model 1 (region exposure):

$$\text{logit } \Pr(y_i = 1) = \eta_i + \beta_R V_{rw}$$

Models 2 and 4 (commune exposure, within and between):

$$\text{logit Pr}(y_i = 1) = \eta_i + \beta_W V_{cw}^{\text{within}} + \beta_B V_c^-$$

Model 3 (same, plus a commune random intercept):

$$\text{logit Pr}(y_i = 1) = \eta_i + \beta_W V_{cw}^{\text{within}} + \beta_B V_c^- + u_c, \quad u_c \sim N(0, \tau^2)$$

- β_W (**within**): when a commune is more violent than its own usual level, is favorability lower? Stable features of the commune (ethnic make-up, remoteness, history) cannot drive this estimate. This is the main quantity.
- β_B (**between**): are people in usually more violent communes less favorable? This compares different communes, so it is more exposed to confounding.

3.3 Estimation

- **M1 and M4** are survey-weighted logistic regressions (`survey::svyglm`). Weights are rescaled within each wave, $w_i^* = w_i n_w / \sum_{i \in w} w_i$, so no wave dominates. Strata are wave-specific. Standard errors use Taylor linearization, clustering on wave-specific **PSUs in M1** and wave-specific **communes in M4**. M4 clusters at the commune because that is where violence is assigned: everyone in a commune-wave shares one violence value.
- **M2** is an ordinary logistic regression (`glm`) that ignores weights and clustering, included only to show how much the clustering correction matters.
- **M3** is a multilevel logistic regression (`lme4::glmer`, Laplace approximation) with a commune random intercept and no weights.

Reported effect. All models report the **average marginal effect** (AME): the average change in the probability of a favorable view, in percentage points, per one-unit increase in log events. Because $\log(1 + E)$ rises by about 0.69 when events double, the table also gives the effect of a **doubling of events**. M3's AME is evaluated at $u_c = 0$ (a typical commune), so it is close to, but not exactly comparable with, the population-averaged AMEs of the other models.

4 Results

Table 3: Violence AMEs in percentage points, with 95% CIs. ‘Per doubling’ multiplies the AME by $\log(2)$, the approximate change in $\log(1 + \text{events})$ when events double. The binary rows compare any event with none, so their doubling column is blank.

Model	Part	AME [95% CI]	SE	Per doubling
M1	Region-wave	6.0 [−4.0, 16.0]	5.1	4.2
M2	Within commune	−9.2 [−12.6, −5.9]	1.7	−6.4
M2	Between communes	−9.4 [−11.8, −6.9]	1.2	−6.5
M3	Within commune	−9.3 [−12.7, −5.9]	1.7	−6.4
M3	Between communes	−9.2 [−11.8, −6.5]	1.3	−6.4
M4	Within commune	−8.1 [−12.9, −3.2]	2.5	−5.6
M4	Between communes	−7.4 [−10.3, −4.4]	1.5	−5.1
M4 + attitudes	Within commune	−5.7 [−10.4, −0.9]	2.4	−3.9
M4 + attitudes	Between communes	−4.9 [−7.8, −1.9]	1.5	−3.4
M4, any event (binary)	Within commune	−8.4 [−14.8, −2.0]	3.2	
M4, any event (binary)	Between communes	−7.1 [−11.5, −2.8]	2.2	

Main result (M4). When a commune experiences more violence than its usual level, favorability changes by **−8.1 points** per unit of log events (95% CI −12.9 to −3.2), or about **−5.6 points per doubling** of events. Comparing communes with each other, the between-commune AME is −7.4 points (95% CI −10.3 to −4.4).

Model 1 for comparison. The region-wave model gives 6.0 points (95% CI −4.0 to 16.0). Its interval is wide because, as Table 2 shows, almost all of its exposure variation is absorbed by the dummies.

Effect of handling clustering. M2 ignores clustering. The M4 standard error for the within effect is 1.4 times larger than in M2. The wider M4 interval is the honest one.

Attitudes. Adding local government approval, support for democracy, perceived violence, and stress (M4 + attitudes) shows whether the association runs partly through these attitudes. Because violence may itself change them, this row is descriptive, not the main estimate.

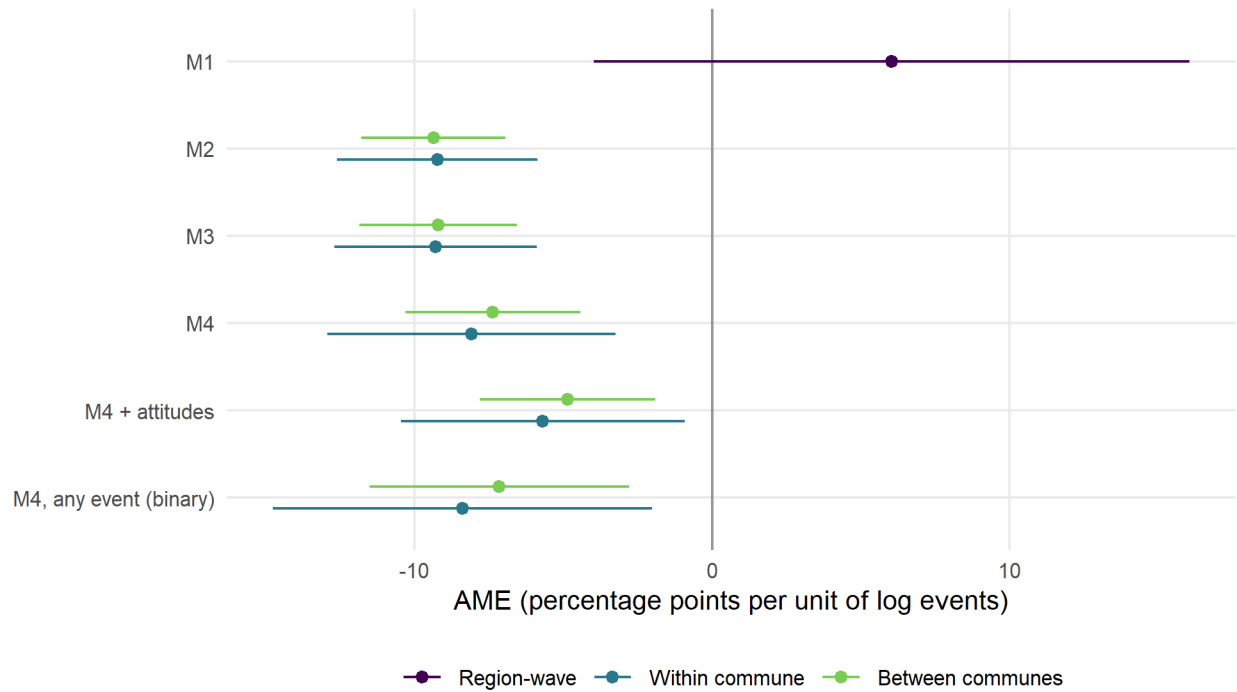


Figure 2: Violence AMEs across models (percentage points, 95% CI). M1 has a single region-wave effect; the commune models split it into within and between.

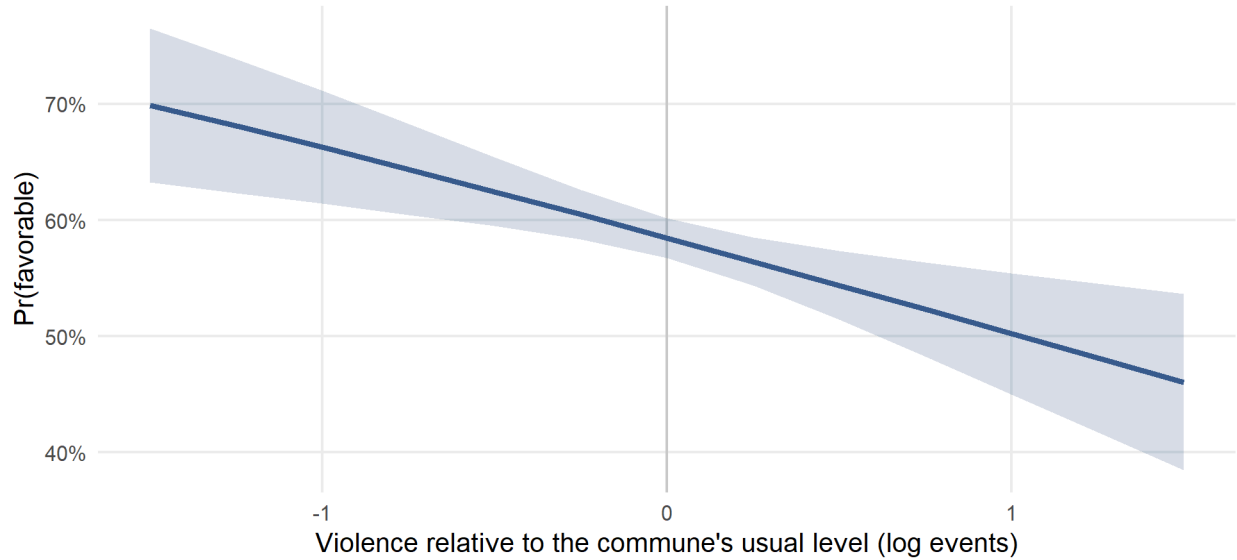


Figure 3: Main model (M4): weighted average predicted probability of a favorable view as a commune’s violence moves above or below its own usual level. +0.69 on the x-axis is roughly a doubling of events.

5 Maps

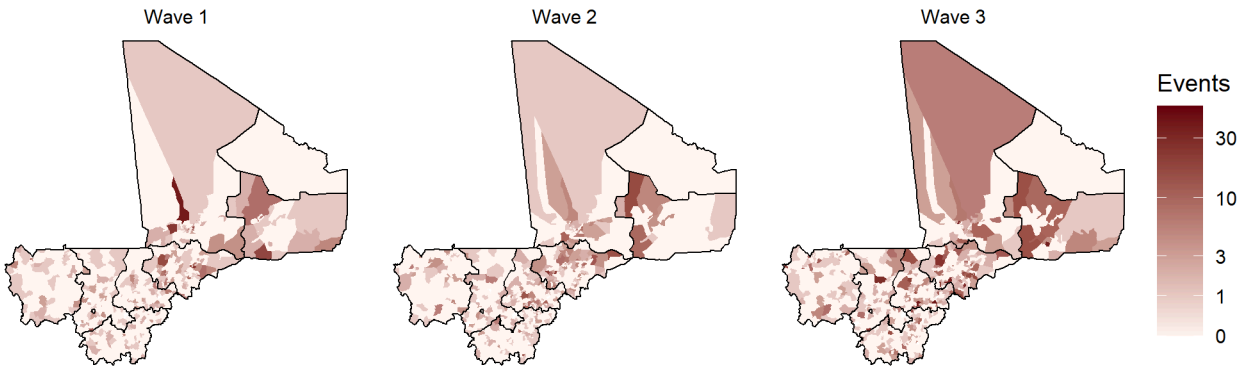


Figure 4: Exposure used by Models 2 to 4: ACLED events by commune and wave. Darker red means more events.

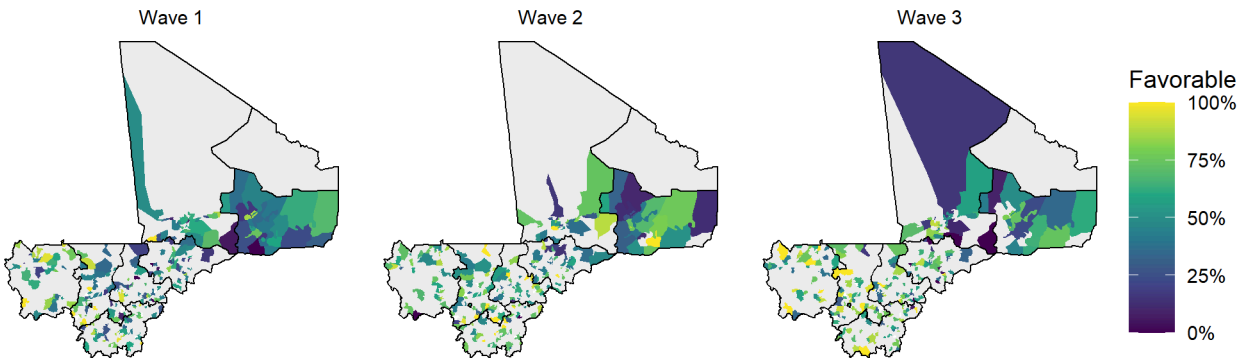


Figure 5: Model 4 view: design-weighted percent favorable by commune and wave (sampled communes only; gray = not sampled that wave). Each value rests on few respondents, so read the map for broad patterns.

Median respondents per commune-wave on the map: 7.

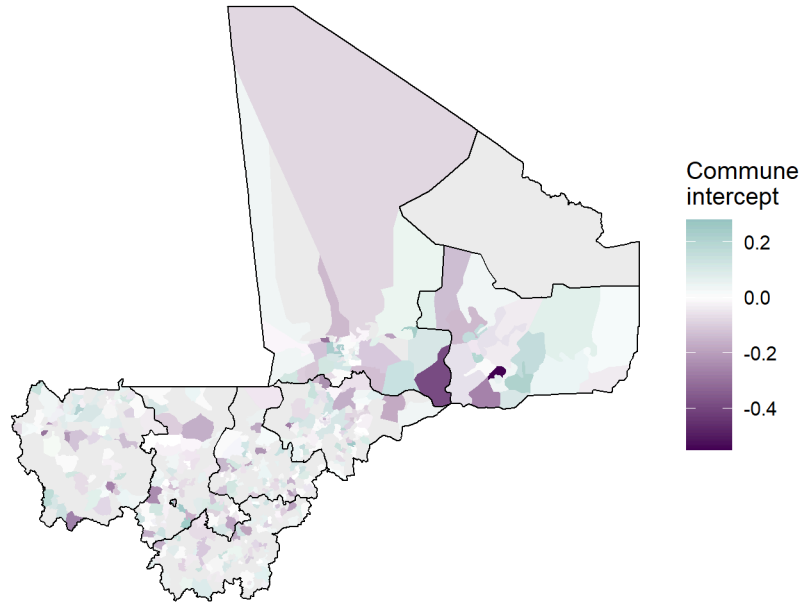


Figure 6: Model 3 view: commune random intercepts. Green communes are more favorable than the model predicts from region, wave, violence, and respondent characteristics; purple are less. Same-colour clusters spanning many communes would suggest a geographic factor the model is missing.

6 Model diagnostics

Model 3 converged without warnings. Its commune variance is $\hat{\tau}^2 = 0.071$ (ICC 0.021 on the latent scale).

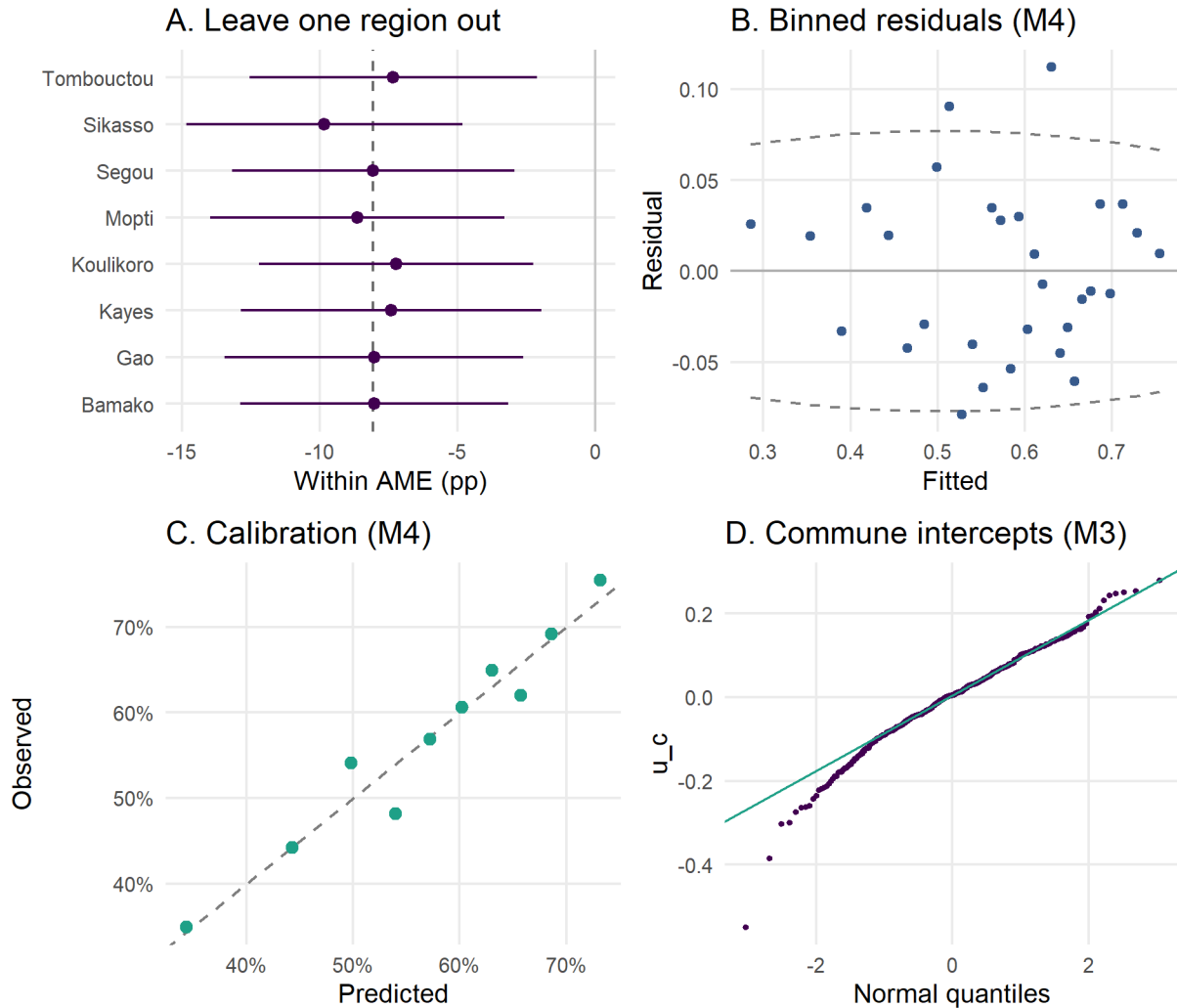


Figure 7: (A) M4 within AME with each region dropped in turn; dashed line = full sample. (B) M4 weighted binned residuals; points should fall inside the dashed bounds with no pattern. (C) M4 calibration by decile. (D) M3 commune intercepts against a normal distribution.

7 Explaining the result

A plain-language version for non-technical audiences:

We compared the same communes across survey rounds. When a commune saw more violent events than it usually does, residents were about 5.6 percentage points less likely to view the leader favorably for each doubling of events, after accounting for region, survey round, and who was interviewed. The range of plausible values is -9.0 to -2.2 points.

8 Limitations

- **Name matching.** Communes are linked by name. A misspelled name silently becomes “0 events”. Table 1 reports names missing from the official commune list; fix those before interpreting results.
- **Communes surveyed once** add nothing to the within estimate (W in Table 2).

- **Serial correlation.** Strata and clusters are wave-specific, so correlation within the same commune across waves is not reflected in M4's standard errors. A conservative check clusters on commune across waves with region as the stratum (see MODELING_NOTES.md).
- **Exposure measurement.** ACLED undercounts events in remote areas, which likely weakens the association. The event window must end before each wave's fieldwork.
- **Confounding.** The within estimate removes stable commune differences but not changes over time that coincide with violence (displacement, military operations, local economic shocks).
- **Coverage.** Insecure areas may be harder to survey, so estimates describe the reachable population.

9 References

- Bell, A., & Jones, K. (2015). Explaining fixed effects: Random effects modeling of time-series cross-sectional and panel data. *Political Science Research and Methods*, 3(1), 133-153.
- Lumley, T. (2010). *Complex surveys: A guide to analysis using R*. Wiley.
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- Raleigh, C., Linke, A., Hegre, H., & Karlsen, J. (2010). Introducing ACLED: An armed conflict location and event dataset. *Journal of Peace Research*, 47(5), 651-660.
- Runfola, D., et al. (2020). geoBoundaries: A global database of political administrative boundaries. *PLoS ONE*, 15(4), e0231866. <https://doi.org/10.1371/journal.pone.0231866>